Football Loans: Empirical Progress

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1 Loan-ending timing

What we are trying to identify

We want to distinguish loan endings that occur on recurring, "normal" country-specific dates from endings that occur at unusual times.

The classification is a **timing signal only**. An off-calendar ending does not by itself prove early termination, recall, option exercise, or any other contractual mechanism.

Data and calendar construction

There are:

- 29,699 loan episodes;
- 29,683 recorded loan endings;
- 27,402 realised loan endings used for the calendar analysis.

The first version of the analysis had poor geographic coverage. Club countries were recovered by revisiting the upstream Transfermarkt histories and extracting the country associated with each club's flag. This information was already present in the source histories but had not been carried through consistently to the loan-level dataset, leaving many endings without a country assignment in the first version. Mapping these flags back to clubs increased country coverage to 27,397 of 27,402 realised endings (99.98%). After recovering club countries primarily from the country flags already contained in the upstream Transfermarkt histories:

	Calendar v1	Calendar v2
Realised endings	27,402	27,402
Classifiable	12,786	25,818
Classifiable share	46.7%	94.2%

Both clubs now have a country for 27,397 of 27,402 realised endings (99.98%).

For each country, we learn recurring ending dates from non-COVID domestic loan endings:

- at least 100 historical endings;
- dates clustered within ±3 days;

- a date must recur in at least 3 calendar years;
- a cluster must account for at least 2% of endings;
- retain major dates until approximately 90% coverage, subject to a maximum of 5 dates.

Cross-border loans use a pooled cross-border calendar.

Although country coverage reaches 99.98%, not every ending is classifiable: classification additionally requires an applicable country-specific (or pooled cross-border) calendar satisfying the historical sample and recurrence thresholds above. This leaves 25,818 of 27,402 realised endings (94.2%) classifiable.

Nearly all unclassifiable endings (1,579) involve countries with insufficient domestic loan history to estimate a reliable country-specific calendar; only 5 are due to unresolved club geography. Some of the largest below-threshold countries are:

Country	**Affected endings**
Israel	98
Thailand	78
Costa Rica	64
Uruguay	64
Ecuador	60
Hungary	57
Qatar	56
Vietnam	53
Slovakia	51
South Africa	49

Thus, *unclassifiable* does not mean that an ending occurred at an unusual time. It means that there is insufficient historical information to construct an applicable calendar against which to classify its timing.

Examples

Calendar	Main recurring ending dates
Spain	Jun 30; Jan 30
Italy	Jun 30; Jan 30
France	Jun 30
Germany	Jun 30; Jan 30
Portugal	Jun 30; Dec 31
England	May 31; Jan 1; Jun 30; Jan 5; Feb 1
Scotland	May 31; Jan 1; May 5; May 1; May 16
Cross-border	Jun 30; Dec 31; Jan 30; May 31

These are **empirical loan-ending dates**, not official transfer-window dates.

Main result

Among the 25,818 classifiable endings:

$$21,300\ (82.5\%) \quad \text{are calendar-normal}$$

and

$$4,518\ (17.5\%) \quad \text{are off-calendar.}$$

Outside COVID, the off-calendar share is 15.2%. During COVID it is 44.4%.

The two groups look quite different:

	Normal	Off-calendar
N	21,300	4,518
Median duration	284 days	164 days
January ending	6.2%	25.1%
July/August ending	2.1%	27.0%
Existing early-termination flag	2.3%	23.9%
Next move $\leq$ 1 day	23.4%	26.6%
Next move $\leq$ 21 days	37.6%	42.9%
Next move $\leq$ 60 days	57.2%	53.8%
Permanent move to borrower	15.5%	11.0%
Loan to different club	35.3%	48.3%

Interpretation. Off-calendar endings are shorter and are much more likely to occur mid-season and to be followed by a loan elsewhere. They look meaningfully different, but the timing alone cannot tell us why the loan ended.

Timing + subsequent move

We also combine calendar status with whether another move occurs within 60 days.

	N	Share
Normal ending, no move $\leq$ 60d	9,283	36.0%
Normal ending, move $\leq$ 60d	11,998	46.5%
Off-calendar, no move $\leq$ 60d	2,088	8.1%
Off-calendar, move $\leq$ 60d	2,412	9.4%

The last group is probably the highest-priority group for future contract-term search because both the ending date and subsequent movement look unusual.

Again, "quick subsequent move" does not mean "planned" and "off-calendar" does not mean "early termination."

Next step: We can examine what types of moves occur after these off-calendar endings (e.g., permanent conversion, re-loan, or loan to a different club) and compare their frequencies and timing with calendar-normal endings. This may help identify which off-calendar cases are genuinely unusual rather than simply reflecting normal registration patterns.

2 What happens after the loan ends?

Our previous broad "another move after return" category mixed together several economically different outcomes.

The old category contained 7,853 cases:

Subsequent pathway	N	Within 1 day
Re-loan to same borrower	331	12%
Loan to different club	5,976	32%
Free move to third club	1,090	52%
No-fee third-club move	382	94%
Other	74	35%

Median time to the subsequent movement also differs:

- same-borrower re-loan: **31 days**;
- different-club loan: **12 days**;
- free third-club move: **1 day**;
- no-fee third-club move: **1 day**;
- other: **8 days**.

Main point. A "return followed by another move" is not one economic event. Most of this old category consists of loans to a *different* club, not repeat loans to the same borrower.

A simple 60-day subsequent-move rule therefore identifies registration sequences but does not identify the underlying contractual mechanism.

3 Playing time: loans versus permanent transfers

Sample and outcome

The analysis covers 14 long-covered top divisions:

Belgium, Denmark, Spain, France, England, Greece, Italy, Germany, Netherlands, Portugal, Russia, Scotland, Türkiye, Ukraine.

Seasons run from 2012/13 through 2025/26.

The revised sample contains:

$$5,868 \text{ loan episodes} \quad \text{and} \quad 14,478 \text{ permanent-transfer episodes.}$$

We removed the earlier requirement that two-thirds of the season remain after arrival. That condition unnecessarily excluded January transfers.

Instead, playing time is already normalized by the player's actual opportunity window:

$$PT_i = \frac{\text{league minutes played after joining}}{90 \times \text{receiving-club league matches while the player is at the club}}.$$

Therefore:

- if a player joins in January, only games after his arrival count;
- if he leaves early, games after departure do not count.

The outcome can be read as:

> the fraction of league minutes the player could have played after joining that he actually played.

Age composition

The age distributions are very similar:

	Loan	Permanent
N	5,868	14,478
Median age	23.1	23.5
Mean age	23.6	24.0

So the results are not simply because loan players are dramatically younger.

Raw age pattern

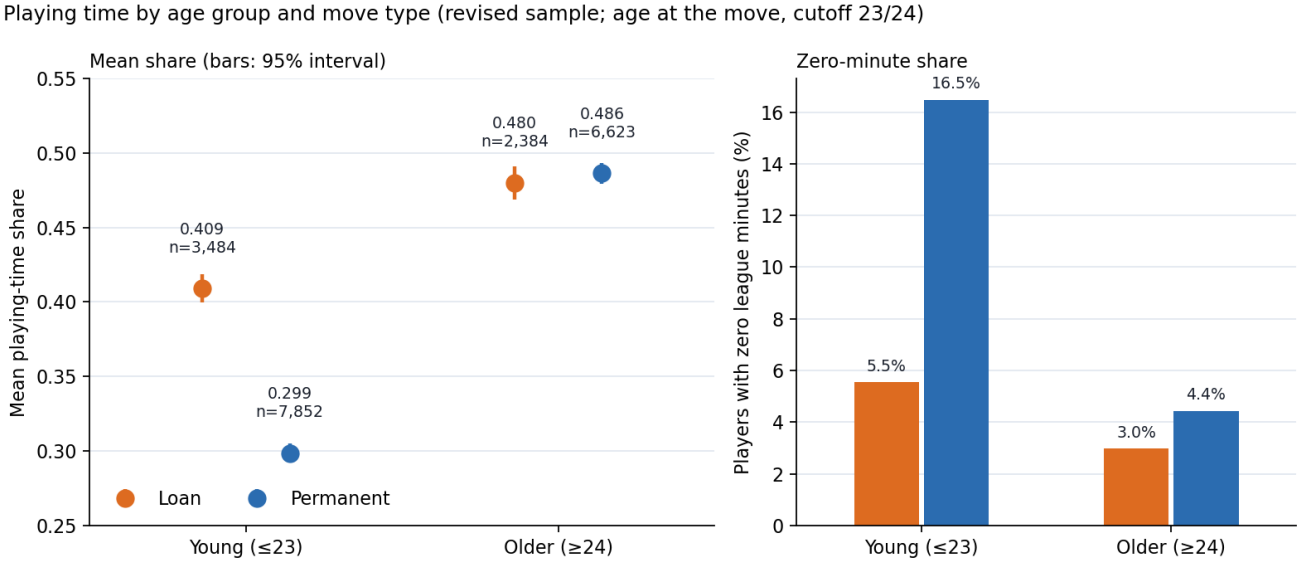
Using young $=\leq 23$ and older $=\geq 24$:

	Loan	Permanent	Raw difference
Young (≤ 23)	0.409	0.299	+11.0 pp
Older (≥ 24)	0.480	0.486	-0.7 pp

Zero playing time is also much more common among young permanent signings:

- young loans: 5.5%;
- young permanent: 16.5%;
- older loans: 3.0%;
- older permanent: 4.4%.

So the raw data suggest a very large playing-time advantage for young loanees, but almost no difference among older players.



Controls

The preferred models control for:

- arrival timing;
- age;
- player value relative to the receiving squad;
- receiving-club squad value;
- position;
- league × season fixed effects;

with standard errors clustered by receiving club.

Relative player value is:

$$RelativeValue_i = \frac{\text{player market value}}{\text{receiving-club squad value}}.$$

This is important because an equally valued player can be central to a small club but marginal at a very expensive squad.

Age result after controls

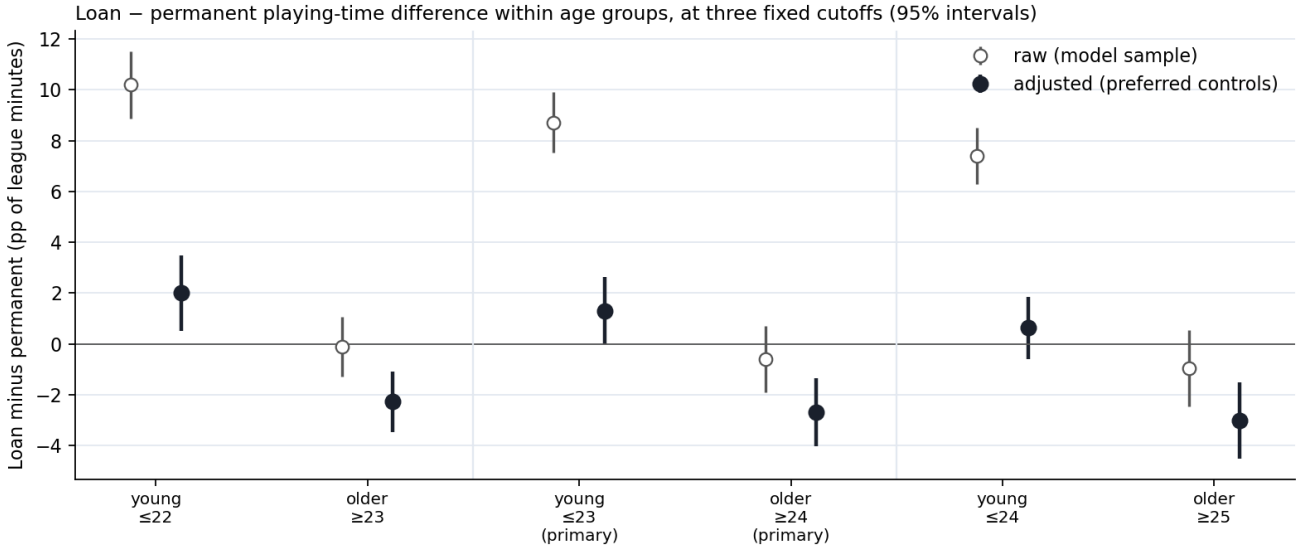
Using the full revised age analysis:

- young loan-minus-permanent difference: approximately **+1.3 pp**;
- older difference: approximately **-2.7 pp**;
- loan × older interaction: approximately **-4.0 pp**.

The age interaction remains similar under neighboring cutoffs:

Age split	Loan × older
≤ 22 versus ≥ 23	-4.3 pp
≤ 23 versus ≥ 24	-4.0 pp
≤ 24 versus ≥ 25	-3.7 pp

Interpretation. The loan-minus-permanent difference is substantially more positive for younger players. Among older players it becomes negative.



Why does the large young-player raw gap disappear?

Market value explains a large part of it.

For young players:

- without the two market-value controls, the loan advantage remains about **+9.1 pp**;
- after controlling for relative player value and squad value, it falls to about **+1.3 pp**.

Median relative value:

- young permanent signing: approximately **0.016**;
- young loanee: approximately **0.034**.

Young loanees are therefore, on average, substantially more valuable relative to the squads that receive them.

This suggests that much of the raw playing-time advantage reflects **selection**: clubs tend to borrow young players whom they expect to be important members of the squad.

England

England is an important exception to the pooled pattern.

In the common regression sample used below:

$$\widehat{\beta}_{Loan,England} = -6.4 \text{ pp}$$

with a 95% confidence interval of approximately

$$[-10.3 \text{ pp}, -2.6 \text{ pp}].$$

So comparable loan players in England receive materially less playing time than permanent signings.

The league heterogeneity reinforces the idea that there is not one universal European "loan effect."

4 Injury robustness

What we did

We integrated an external player-level injury-history dataset into the playing-time calculation.

The key question is:

> If a player was medically unavailable for some of the receiving club's league matches, does removing those games from his opportunity denominator change the loan-versus-permanent result?

The external injury data contain 143,195 raw rows and use Transfermarkt player IDs.

The revised playing-time sample contains 20,346 transfer episodes (5,868 loans and 14,478 permanent transfers), corresponding to 10,188 unique players. We matched these players to the external injury dataset using exact Transfermarkt player IDs. Of the 10,188 players, 8,804 (86.4%) were found in the external dataset: 7,163 had at least one recorded injury and 1,641 had a player profile but no recorded injury. The remaining 1,384 players were not found in the external dataset.

- 7,163 appear in the external data with injury rows;
- 1,641 appear in the player profiles with zero injury rows;
- 1,384 are absent;
- exact-ID coverage is therefore **86.4%**.

For matched players with birth dates on both sides, 99.8% of dates of birth agree.

How the adjustment works

The original outcome is:

$$PT_i = \frac{Minutes_i}{90 \times Games_i}.$$

The injury-adjusted version is:

$$PT_i^{fit} = \frac{Minutes_i}{90 \times (Games_i - InjuryGames_i)}.$$

But we do **not** simply use the source's reported "games missed" number.

Instead, each injury spell is overlaid on the actual receiving club's league fixtures.

A match is removed only when:

1. its date falls inside a recorded medical injury or illness spell;
2. the player does not appear in the match.

If the player appears, the game stays in the denominator even if the injury source says the spell covered that date.

Example: Waldemar Anton

Waldemar Anton moved permanently from Stuttgart to Dortmund in 2024/25.

- 34 Bundesliga games in his observation window;
- 1,948 minutes played.

Original share:

$$\frac{1,948}{90 \times 34} = 0.637.$$

His recorded "thigh problems" spell begins on October 26, but he plays 45 minutes against Augsburg that day. Because he appears, that game stays.

Five later games are covered by recorded medical spells and he does not play:

- Leipzig;
- Mainz;
- Gladbach;
- Hoffenheim;
- Leverkusen.

Therefore:

$$34 - 5 = 29$$

available games after injury adjustment, and

$$\frac{1,948}{90 \times 29} = 0.746.$$

So his playing-time share changes:

$$0.637 \rightarrow 0.746.$$

This example captures the main logic: match-level overlap, confirmed absence, and appearance override.

Coverage and data quality

Usable episodes:

	Usable	Revised sample
Loan	4,526	5,868
Permanent	10,600	14,478

Overall, about 74% of revised episodes are usable for the injury adjustment.

Main limitations:

- the injury snapshot ends on September 15, 2025, so 2025/26 is incomplete;
- 1,772 players hit a 15-injury history cap, so older injuries may be truncated;
- coverage is weaker before 2017.

This is why injury adjustment is a robustness check rather than the primary outcome.

Validation

Because the external injury data are incomplete, we first check how well they capture injury events that we can independently observe in our match data.

Our match data contain 4,976 cases in which a player was substituted off and the substitution was recorded as injury-related. For each of these events, we take the date of the match and ask whether the external injury dataset records an injury for the same player beginning around that date.

Across all 4,976 injury substitutions, the external data record an injury beginning:

- on exactly the same day in 17.1% of cases;
- within ± 3 days in 32.5% of cases;
- within ± 7 days in 34.6% of cases.

However, an injury substitution does not necessarily mean that the player was unavailable for subsequent matches. A player may leave a match with a minor knock and recover in time to play the following game. Such an event may therefore appear as an injury substitution in our match data without generating a meaningful absence spell in the external injury data.

We therefore use a stronger validation test. We focus on injury substitutions after which the player does not appear in either of his club's next two league matches. These cases provide stronger evidence that the injury actually caused match unavailability, which is the type of injury our playing-time adjustment is intended to capture.

For these clearer injury absences, the external dataset records an injury beginning within ± 7 days of the injury substitution in:

- **70.8%** of cases overall;
- **49.3%** of cases from 2012/13 through 2016/17;
- **74.0%** of cases from 2017/18 through 2024/25.

Thus, when our match data provide relatively strong evidence that an injury led to subsequent match absences, the external source identifies a corresponding injury in roughly seven out of ten cases. Coverage is substantially weaker in the earlier seasons and improves considerably from 2017/18 onward.

This validation suggests that the external injury data contain useful information about periods of medical unavailability, particularly in the later seasons, but they are not complete. A missing injury record therefore cannot be interpreted as definitive evidence that a player was healthy. For this reason, we use the injury-adjusted playing-time measure as a robustness check rather than as our primary playing-time outcome.

Does injury adjustment change the result?

The preferred model is run twice on exactly the same 13,967 observations: (The sample falls from 15,126 injury-usable episodes to 13,967 observations because the regression also requires complete control variables. Most of this reduction comes from missing market-value information needed to construct relative player value (player market value relative to the receiving club's squad value.)

	Current	Injury-adjusted	Change
Pooled loan coefficient	+0.5 pp	+0.8 pp	+0.3 pp
Young (≤ 23)	+2.0 pp	+2.3 pp	+0.4 pp
Older (≥ 24)	-2.6 pp	-2.5 pp	+0.2 pp
Loan $\times$ older	-4.6 pp	-4.8 pp	-0.2 pp
England	-6.4 pp	-6.5 pp	≈ 0

For the pooled coefficient:

$$+0.5 \text{ pp} \quad [-0.7, +1.6]$$

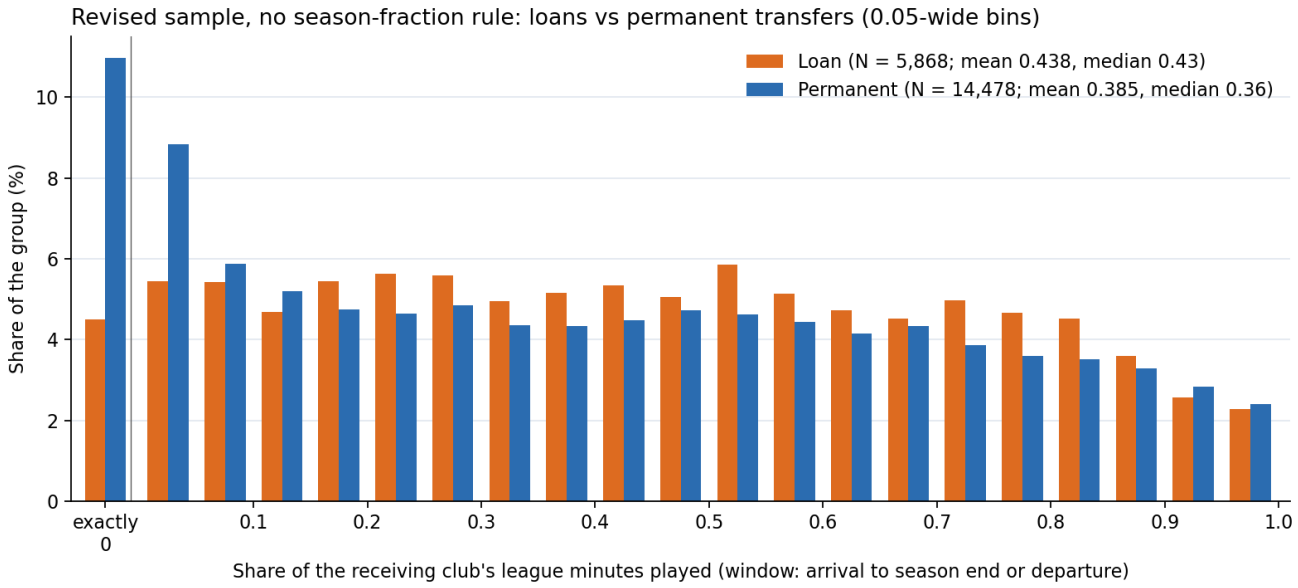
becomes

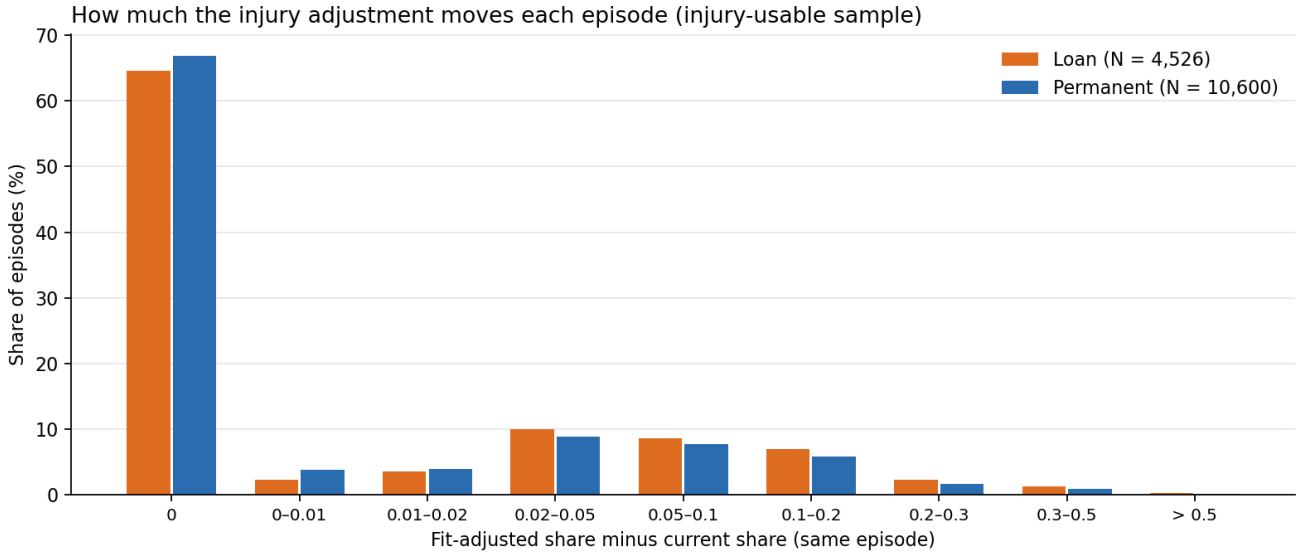
$$+0.8 \text{ pp} \quad [-0.4, +2.0].$$

Alternative injury definitions give essentially the same result:

- strict definition: +0.7 pp;
- only injuries removing at least 3 games: +0.7 pp;
- only injuries removing at least 5 games: +0.7 pp.

Conclusion: injury adjustment does not materially alter the main playing-time results.





5 Where we are now

The four pieces together give the following picture.

1. **Timing is informative.**

   Most loans end on recurring country-specific dates. Off-calendar endings form a meaningful minority and look systematically different, especially in duration and subsequent loan behavior.

2. **The next move matters.**

   A loan return followed by another registration cannot be treated as one outcome. A repeat loan, a loan elsewhere, and a permanent sale are economically different pathways.

3. **There is no single loan-playing-time effect.**

   Young loan players look much better in the raw data, but much of that difference is explained by the fact that they are relatively valuable players for the clubs borrowing them.

   After controls, younger players retain a small positive loan-minus-permanent difference, while older players have a negative difference.

   England is particularly negative.

4. **Injuries do not explain these patterns.**

   Removing confirmed injury games changes individual playing-time measures, sometimes substantially, but barely changes the regression estimates.

6 Questions / next steps

The main next step is to connect these descriptive patterns to actual contract terms.

In particular:

- use off-calendar endings and subsequent-move pathways to prioritize unusual loan episodes;
- search for disclosed recall clauses, purchase options, obligations, permanent conversions, and related terms;
- examine how actual contractual structure relates to the movement and playing-time patterns already identified.

One optional remaining timing sensitivity is to vary the maximum number of calendar dates retained for countries such as England, Scotland, and Brazil. Otherwise, the four analyses above are ready to serve as the empirical foundation for the contract-term stage.